



PROJECT TITLE

For

Sensor Lab mini-Project

Third Year (Semester-VI) Bachelors in Information Technology, Autonomous
Program

By

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Certificate

This is to certify that **Palak,Sonam,Mahi** have completed the project report on the topic **RFID Based Door Lock System** satisfactorily in partial fulfilment of the requirements for the award of Mini Project in sensor lab of third Year, (Semester-VI) in Information Technology under the guidance of **Mr. Abhishek Chaudhari** during the year 2023-2024 as prescribed by An Autonomous Institute Affiliated to University of Mumbai

Supervisor/ Examiner

Abstract

This project focuses on the development of a **secure, contactless RFID-based door lock system** utilizing the **Arduino Mega** microcontroller, an **MFRC522 RFID reader**, and a **servo motor** to control physical access. Designed as a smart, low-cost alternative to traditional locks, the system reads unique RFID card IDs, verifies them against stored credentials, and either grants or denies access by rotating a servo motor to open or lock the door. The solution may also integrate an **I2C LCD display** to show real-time status messages or a **buzzer** to provide auditory feedback. This embedded system enhances convenience and security, making it ideal for residential buildings, offices, and laboratory environments. Additionally, it highlights the potential of RFID and Arduino technology in modern automation and access control systems.

Chapter 1: Introduction

RFID (Radio Frequency Identification) technology is rapidly transforming traditional access control systems by offering a contactless, secure, and efficient solution. In conventional methods such as physical locks or keypad-based systems, there is always a risk of key loss, duplication, or password leakage. These traditional systems also require physical interaction, which can be inconvenient in high-traffic areas. In contrast, RFID-based systems ensure faster authentication, minimal human contact, and centralized management of access credentials, making them ideal for modern automation needs.

This project presents an RFID-based door lock system built using Arduino Mega and the MFRC522 RFID module. The system is designed to scan RFID tags or cards and validate them using a pre-stored UID database. If the scanned card is authorized, the servo motor rotates to unlock the door, and optionally, an LCD display or buzzer provides visual or audio feedback for successful or denied access. After a short delay, the system automatically re-locks the door, enhancing both convenience and security.

With simple hardware and low cost, this project delivers a reliable, scalable, and easy-to-use solution for securing homes, offices, laboratories, and other restricted areas. It also demonstrates the capabilities of embedded systems and RFID technology in IoT and smart security applications.

Chapter 2: Literature Review

2.1 Literature Survey

Radio Frequency Identification (RFID) technology has become a popular solution for secure access control systems. Its ability to operate contactlessly, reliably, and quickly has led to its adoption in homes, offices, transport systems, and industrial automation. An RFID system typically includes a reader module (such as the RC522), RFID tags or cards, a microcontroller (e.g., Arduino Uno or Mega), and an output unit such as a lock, buzzer, or LCD. Academic and industrial research over the past decade has proven the effectiveness of RFID as a cost-effective alternative to physical keys and PIN-based systems.

RFID-based access systems have been integrated with various other components to improve user experience and enhance security. Systems often include features such as LCD displays for status updates, buzzer alerts, and mobile integration. The wide availability and low cost of Arduino-based systems have encouraged students and hobbyists to develop simple prototypes that showcase the power of embedded electronics.

2.2 Related Work

Several RFID-based security systems have been proposed in past studies and DIY implementations. For example, some systems integrate RFID with GSM modules to send alerts via SMS when access is attempted. Other designs involve the use of keypads alongside RFID to implement two-factor authentication. In some commercial systems, biometrics (fingerprint or facial recognition) are combined with RFID for high-security areas like data centers or server rooms.

However, these solutions often come with drawbacks like increased complexity, higher power consumption, and cost escalation. Many of these models, while functional, are not suitable for small-scale applications like student labs, hostels, or home automation due to their bulky nature and overuse of hardware.

2.3 Research Gap

Most existing RFID systems rely on additional hardware such as keypads, biometric scanners, or GSM modules. While they add features, they also make the system more complex and expensive. Moreover, in many cases, UID management is hardcoded into the microcontroller, which means changing or updating authorized cards requires re-uploading the code.

In low-cost academic or DIY setups, systems often lack EEPROM-based UID storage, real-time logging, and network connectivity for remote access. Another gap is the absence of a flexible user interface — either visual (LCD) or audio (buzzer) — which can enhance usability without requiring complex circuits. Many basic implementations also omit automatic relocking mechanisms, which are essential for security.

2.4 Proposed Contribution

This project presents a simplified and efficient RFID-based door lock system using Arduino Uno, RC522 RFID reader, a servo motor, and optionally an I2C LCD or buzzer. The design removes the need for keypads and biometric sensors, offering a cleaner and more beginner-friendly system while still being functional and secure. The system displays a real-time prompt to “Scan Your Card,” validates the UID, unlocks the servo if matched, and relocks after a delay. Unauthorized cards receive an “Access Denied” message.

The design supports future enhancements like UID logging via serial, EEPROM storage for saving multiple authorized UIDs, and expansion into IoT platforms using Wi-Fi/Bluetooth modules. Its strength lies in its simplicity, cost-efficiency, and practical usability, making it ideal for educational labs, home security, and DIY learning projects.

Chapter 3: Design Procedure

3.1 Problem Statement

The primary objective of this project is to design and implement an RFID-controlled door locking system using an Arduino Uno, eliminating the need for traditional keys or keypad input. The system ensures secure, contactless access through RFID card scanning and mechanical door actuation using a servo motor. It is especially suited for labs, smart homes, and institutional entry systems where a low-cost, simple yet secure access mechanism is required. This system provides real-time feedback via a buzzer or LCD and automatically re-locks the door after a delay, increasing both security and convenience.

3.2 Steps Involved in Project Development

3.2.1 Component Selection and Procurement

The components were selected based on compatibility, simplicity, cost-effectiveness, and ease of availability. Below are the key components:

- Arduino Uno: Acts as the central controller, managing the RFID module, servo, and output components.
- MFRC522 RFID Reader: Used to scan RFID cards or tags; communicates with Arduino via SPI protocol.
- RFID Tags/Cards: Passive cards that contain a unique identifier (UID) for authentication.
- Servo Motor: Used as the locking/unlocking mechanism; controlled by PWM signals from the Arduino.
- I2C LCD Display (Optional): Displays messages such as "Access Granted" or "Access Denied".
- Buzzer (Optional): Provides audio feedback to the user.
- Power Supply: 5V/9V battery or USB connection to power the Arduino and other components.
- Breadboard and Jumper Wires: For connecting and prototyping the components.

3.2.2 Circuit Assembly and Wiring

In this phase, the components are connected as per the defined pin mappings:

- The RFID Reader (RC522) is connected to the Arduino Uno using SPI interface (SDA to Pin 10, SCK to Pin 13, MOSI to Pin 11, MISO to Pin 12, RST to Pin 9).
- The Servo Motor is connected to digital Pin 3 of Arduino.
- LCD (if used) is connected via I2C (SDA to A4, SCL to A5).
- The Buzzer is optionally connected to Pin 4.
- All components share a common GND and are powered from Arduino's 5V output.

3.2.3 Arduino Programming

The Arduino is programmed using the Arduino IDE. The software workflow involves:

- Initializing the RFID module, Servo motor, and optional LCD/buzzer.
- Continuously monitoring for new RFID cards.
- Reading the card UID and comparing it with a pre-defined authorized UID.
- If the UID matches:
 - Rotate the servo to unlock.
 - Display “Access Granted” or beep.
 - After a delay, lock again.
- If the UID does not match:
 - Display “Access Denied” or beep differently.

The logic ensures a smooth and automatic re-locking mechanism.

3.2.4 Testing and Calibration

Testing involved scanning both valid and invalid RFID cards. Calibration steps included:

- Setting correct angle of servo motor for locked (0°) and unlocked (90°) positions.
- Verifying real-time responses of LCD and buzzer.
- Adjusting delay timing for auto-lock (usually set to 3–5 seconds).
- Ensuring stable power supply and proper card read range (2–3 cm).

The testing process confirmed the system’s reliability and response time.

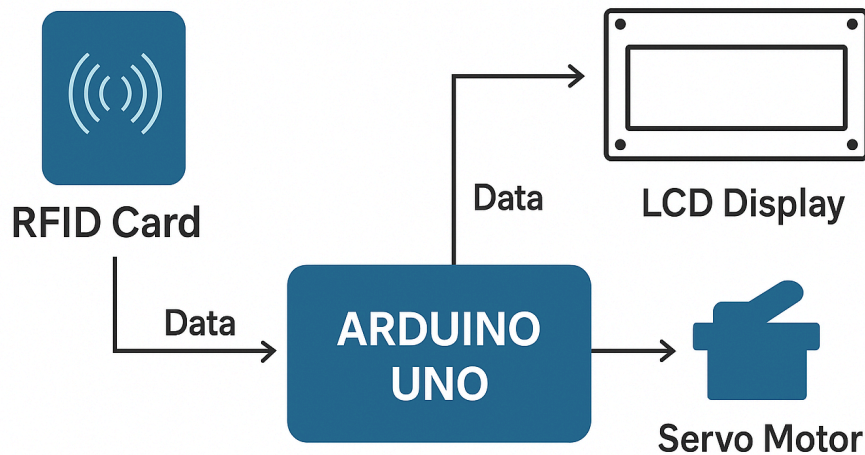
3.2.5 Final Assembly and Demonstration

After successful circuit-level testing and software verification, all components were mounted properly on a cardboard/mock door model. The final demonstration involved:

- Powering the system via USB or battery.
- Displaying “Scan Your Card” on LCD.
- Unlocking and relocking via servo on valid scan.
- Showing “Access Denied” on LCD or buzzer on invalid scan.

This demonstration validated the effectiveness of the RFID-based access system.

3.3 Block Diagram of Proposed System



RFID-Based Door Lock System

3.4 Component Description

3.4.1 Hardware Components

- Arduino Uno: Microcontroller responsible for handling RFID input and controlling the servo, buzzer, and LCD.
- MFRC522 RFID Reader: Module used to read RFID tags operating at 13.56 MHz.
- RFID Tags/Cards: Contains a unique ID (UID) used for authentication.
- Servo Motor (SG90): Provides mechanical movement to lock/unlock mechanism.
- I2C LCD (Optional): Displays system messages.
- Buzzer (Optional): Provides sound feedback.
- Breadboard and Jumper Wires: Used for prototyping connections.
- Power Supply: USB cable or external 5V/9V source.

3.4.2 Specifications

- Operating Voltage: 5V DC
- RFID Operating Frequency: 13.56 MHz
- Servo Range: 0° to 180° (used 0° for lock, 90° for unlock)
- Read Distance: 2–5 cm (RFID)
- Microcontroller Clock Speed: 16 MHz (Arduino Uno)

3.5 Working of the Proposed Project

The RFID card contains a unique ID, which is read by the MFRC522 module when brought near. This UID is sent to the Arduino Uno, which compares it with a stored authorized UID. If matched, the servo motor rotates to unlock the door and the LCD shows "Access Granted." If not matched, the LCD shows "Access Denied" and the buzzer gives a different alert tone. After a predefined delay, the servo returns to its initial (locked) position.

The entire process is automated and runs continuously, waiting for the next card scan. The project demonstrates an efficient and secure system for real-time access control without requiring a keypad, making it ideal for practical and educational applications.

Chapter 4: Implementation

Hardware Implementation:

- Connect RFID RC522 using SPI pins (10, 9, 50-53)
- Connect Servo to Pin 3
- Connect Buzzer to Pin 4 (optional)

Software Implementation:

```
#include <SPI.h>
#include <MFRC522.h>
#include <Wire.h>
#include <LiquidCrystal_I2C.h>
#include <Servo.h>
```

```
#define SS_PIN 10
#define RST_PIN 9
MFRC522 rfid(SS_PIN, RST_PIN);
```

```
LiquidCrystal_I2C lcd(0x27, 16, 2);
Servo lockServo;
```

```
byte authorizedUID[] = {0xA1, 0xB2, 0xC3, 0xD4}; // Example UID
```

```
void setup() {
  Serial.begin(9600);
  SPI.begin();
  rfid.PCD_Init();
```

```
  lcd.begin();
```

```

    lcd.backlight();
    lcd.setCursor(0, 0);
    lcd.print("Scan Your Card");

    lockServo.attach(3);
    lockServo.write(0); // Locked
}

void loop() {
    if (!rfid.PICC_IsNewCardPresent() || !rfid.PICC_ReadCardSerial()) return;

    byte* id = rfid.uid.uidByte;
    bool authorized = true;

    for (byte i = 0; i < rfid.uid.size; i++) {
        if (id[i] != authorizedUID[i]) {
            authorized = false;
            break;
        }
    }

    lcd.clear();
    if (authorized) {
        lcd.setCursor(0, 0);
        lcd.print("Access Granted");
        lockServo.write(90);
        delay(3000);
        lockServo.write(0);
    } else {
        lcd.setCursor(0, 0);
        lcd.print("Access Denied");
        delay(2000);
    }

    lcd.clear();
    lcd.setCursor(0, 0);
    lcd.print("Scan Your Card");

    rfid.PICC_HaltA();
    rfid.PCD_StopCrypto1();
}

```

Chapter 5: Results and Discussion

5.1 Simulation Output

The project was initially tested through the Arduino IDE using the Serial Monitor. The RFID RC522 module correctly scanned the cards, and the UID values were displayed accurately. Upon matching with the pre-defined authorized UID, the system sent control signals to the servo motor, which then rotated to unlock the door. Simultaneously, the LCD displayed “Access Granted” and reverted back to the initial state after a delay. In case of unauthorized cards, the LCD displayed “Access Denied” and no movement occurred in the servo motor. The buzzer provided audio confirmation during both cases. The simulation verified that the logic and code functioned as expected.

5.2 Actual Hardware Output

On actual hardware implementation, the system functioned smoothly. RFID cards were scanned using the MFRC522 reader, and real-time feedback was observed on the LCD. The servo motor unlocked when an authorized UID was detected and automatically re-locked after a 3-second delay. When an unauthorized card was scanned, the system denied access, and feedback was given via both the LCD and the buzzer. The project operated effectively under normal power conditions, and the RFID reader was able to detect cards within a range of 2–4 cm, providing fast and reliable performance.

5.3 Discussion

This project successfully validates the concept of a contactless, RFID-based access control system. The system is low-cost, easy to assemble, and accurate. Below is a breakdown of its strengths and potential improvements:

Pros:

- Fast and secure card authentication
- Contactless and convenient access
- Easy to program and expand
- Reliable servo locking mechanism
- Optional feedback via LCD or buzzer

Areas of Improvement:

- EEPROM support can be added to dynamically store and update multiple authorized UIDs without re-uploading the code.
- Integration with Wi-Fi or GSM modules can provide remote access logs and control.
- Implementation of a real-time clock (RTC) could help in timestamp-based access logging.

References

1. Arduino Documentation:
<https://www.arduino.cc/reference/en/>
2. MFRC522 RFID Reader Datasheet:
<https://www.nxp.com/docs/en/data-sheet/MFRC522.pdf>
3. Servo Motor (SG90) Datasheet:
<https://www.towerpro.com.tw/product/sg90-9g-micro-servo/>

Conclusion:

This project demonstrates the successful implementation of a smart door lock system based on RFID technology. By using an Arduino Uno, MFRC522 RFID reader, and a servo motor, a secure and reliable locking system was created without requiring physical keys or keypads. The system is highly suitable for homes, labs, and office environments, offering enhanced convenience and automation.

Its scalability and modular design allow for future upgrades such as mobile integration, internet connectivity (IoT), and EEPROM-based UID management. Overall, the project highlights how embedded systems and wireless technologies can be combined to solve real-world security problems in a practical and cost-effective manner.